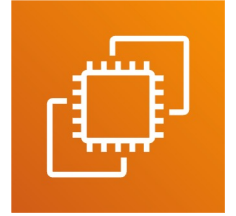


EC2 Section



Amazon EC2

- EC2 is one of the most popular of AWS' offering
- EC2 = Elastic Compute Cloud = Infrastructure as a Service
- It mainly consists in the capability of :
 - Renting virtual machines (EC2)
 - Storing data on virtual drives (EBS)
 - Distributing load across machines (ELB)
 - Scaling the services using an auto-scaling group (ASG)
- Knowing EC2 is fundamental to understand how the Cloud works

EC2 sizing & configuration options

- Operating System (**OS**): Linux, Windows or Mac OS
- How much compute power & cores (**CPU**)
- How much random-access memory (**RAM**)
- How much storage space:
 - Network-attached (**EBS & EFS**)
 - hardware (**EC2 Instance Store**)
- Network card: speed of the card, Public IP address
- Firewall rules: **security group**
- Bootstrap script (configure at first launch): EC2 User Data

EC2 User Data

- It is possible to bootstrap our instances using an [EC2 User data](#) script.
- [bootstrapping](#) means launching commands when a machine starts
- That script is [only run once](#) at the instance [first start](#)
- EC2 user data is used to automate boot tasks such as:
 - Installing updates
 - Installing software
 - Downloading common files from the internet
 - Anything you can think of
- The EC2 User Data Script runs with the root user

Hands-On: Launching an EC2 Instance running Linux

- We'll be launching our first virtual server using the AWS Console
- We'll get a first high-level approach to the various parameters
- We'll see that our web server is launched using EC2 user data
- We'll learn how to start / stop / terminate our instance.

EC2 Instance Types - Overview

- You can use different types of EC2 instances that are optimised for different use cases (<https://aws.amazon.com/ec2/instance-types/>)
- AWS has the following naming convention:

m5.2xlarge

- m: instance class
- 5: generation (AWS improves them over time)
- 2xlarge: size within the instance class

General Purpose

Compute Optimized

Memory Optimized

Accelerated Computing

Storage Optimized

HPC Optimized

Instance Features

Measuring Instance
Performance

EC2 Instance Types – General Purpose

- Great for a diversity of workloads such as web servers or code repositories
- Balance between:
 - Compute
 - Memory
 - Networking
- In the course, we will be using the t2.micro which is a General Purpose EC2 instance

General Purpose

General purpose instances provide a balance of compute, memory and networking resources, and can be used for a variety of diverse workloads. These instances are ideal for applications that use these resources in equal proportions such as web servers and code repositories.

Mac	T4g	T3	T3a	T2	M6g	M5	M5a	M5n	M5zn	M4	A1
-----	-----	----	-----	----	-----	----	-----	-----	------	----	----

** this list will evolve over time, please check the AWS website for the latest information*

EC2 Instance Types – Compute Optimized

- Great for compute-intensive tasks that require high performance processors:
 - Batch processing workloads
 - Media transcoding
 - High performance web servers
 - High performance computing (HPC)
 - Scientific modeling & machine learning
 - Dedicated gaming servers

Compute Optimized

Compute Optimized instances are ideal for compute bound applications that benefit from high performance processors. Instances belonging to this family are well suited for batch processing workloads, media transcoding, high performance web servers, high performance computing (HPC), scientific modeling, dedicated gaming servers and ad server engines, machine learning inference and other compute intensive applications.



** this list will evolve over time, please check the AWS website for the latest information*

EC2 Instance Types – Memory Optimized

- Fast performance for workloads that process large data sets in memory
- Use cases:
 - High performance, relational/non-relational databases
 - Distributed web scale cache stores
 - In-memory databases optimized for BI (business intelligence)
 - Applications performing real-time processing of big unstructured data

Memory Optimized

Memory optimized instances are designed to deliver fast performance for workloads that process large data sets in memory.

R6g	R5	R5a	R5b	R5n	R4	X1e	X1	High Memory	z1d
-----	----	-----	-----	-----	----	-----	----	-------------	-----

** this list will evolve over time, please check the AWS website for the latest information*

EC2 Instance Types – Storage Optimized

- Great for storage-intensive tasks that require high, sequential read and write access to large data sets on local storage
- Use cases:
 - High frequency online transaction processing (OLTP) systems
 - Relational & NoSQL databases
 - Cache for in-memory databases (for example, Redis)
 - Data warehousing applications
 - Distributed file systems

Storage Optimized

Storage optimized instances are designed for workloads that require high, sequential read and write access to very large data sets on local storage. They are optimized to deliver tens of thousands of low-latency, random I/O operations per second (IOPS) to applications.

I3	I3en	D2	D3	D3en	H1
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** this list will evolve over time, please check the AWS website for the latest information*

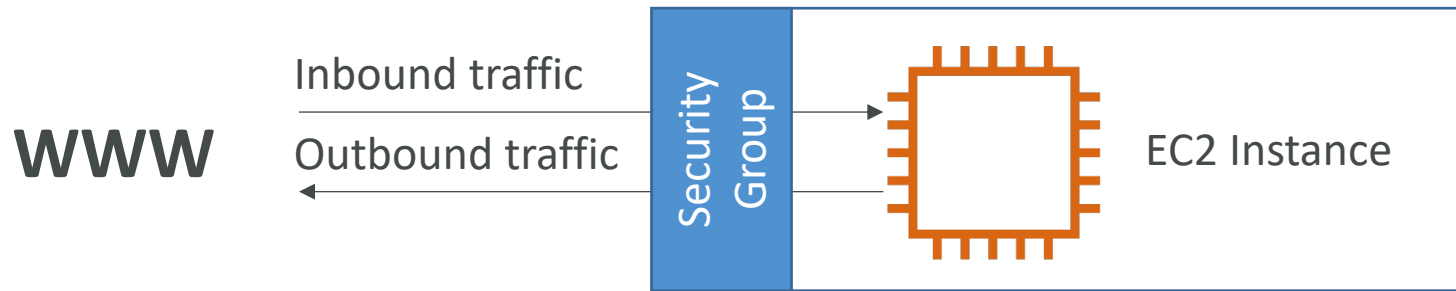
EC2 Instance Types: example

Instance	vCPU	Mem (GiB)	Storage	Network Performance	EBS Bandwidth (Mbps)
t2.micro	1	1	EBS-Only	Low to Moderate	
t2.xlarge	4	16	EBS-Only	Moderate	
c5d.4xlarge	16	32	1 x 400 NVMe SSD	Up to 10 Gbps	4,750
r5.16xlarge	64	512	EBS Only	20 Gbps	13,600
m5.8xlarge	32	128	EBS Only	10 Gbps	6,800

Great website: <https://instances.vantage.sh>

Introduction to Security Groups

- Security Groups are the fundamental of network security in AWS
- They control how traffic is allowed into or out of our EC2 Instances.



- Security groups only contain **allow** rules
- Security groups rules can reference by IP or by security group

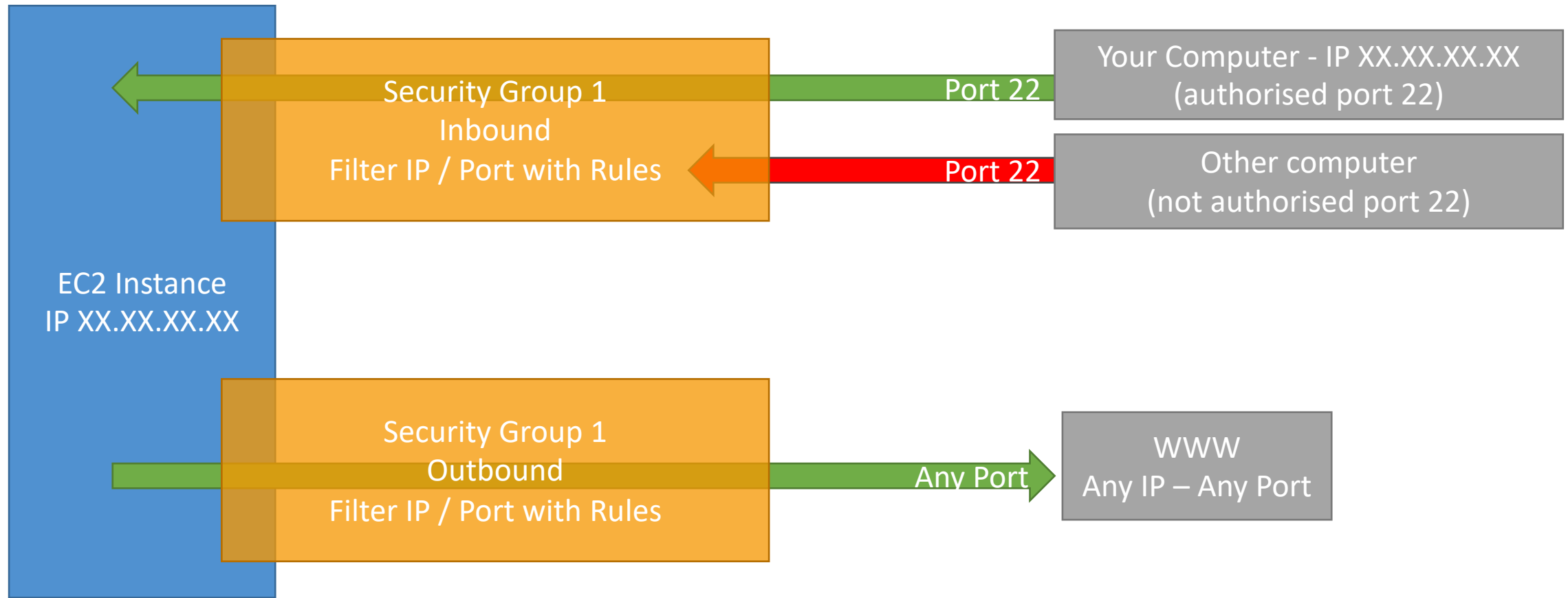
Security Groups

Deeper Dive

- Security groups are acting as a “firewall” on EC2 instances
- They regulate:
 - Access to Ports
 - Authorised IP ranges – IPv4 and IPv6
 - Control of inbound network (from other to the instance)
 - Control of outbound network (from the instance to other)

Type ⓘ	Protocol ⓘ	Port Range ⓘ	Source ⓘ	Description ⓘ
HTTP	TCP	80	0.0.0.0/0	test http page
SSH	TCP	22	122.149.196.85/32	
Custom TCP Rule	TCP	4567	0.0.0.0/0	java app

Security Groups Diagram



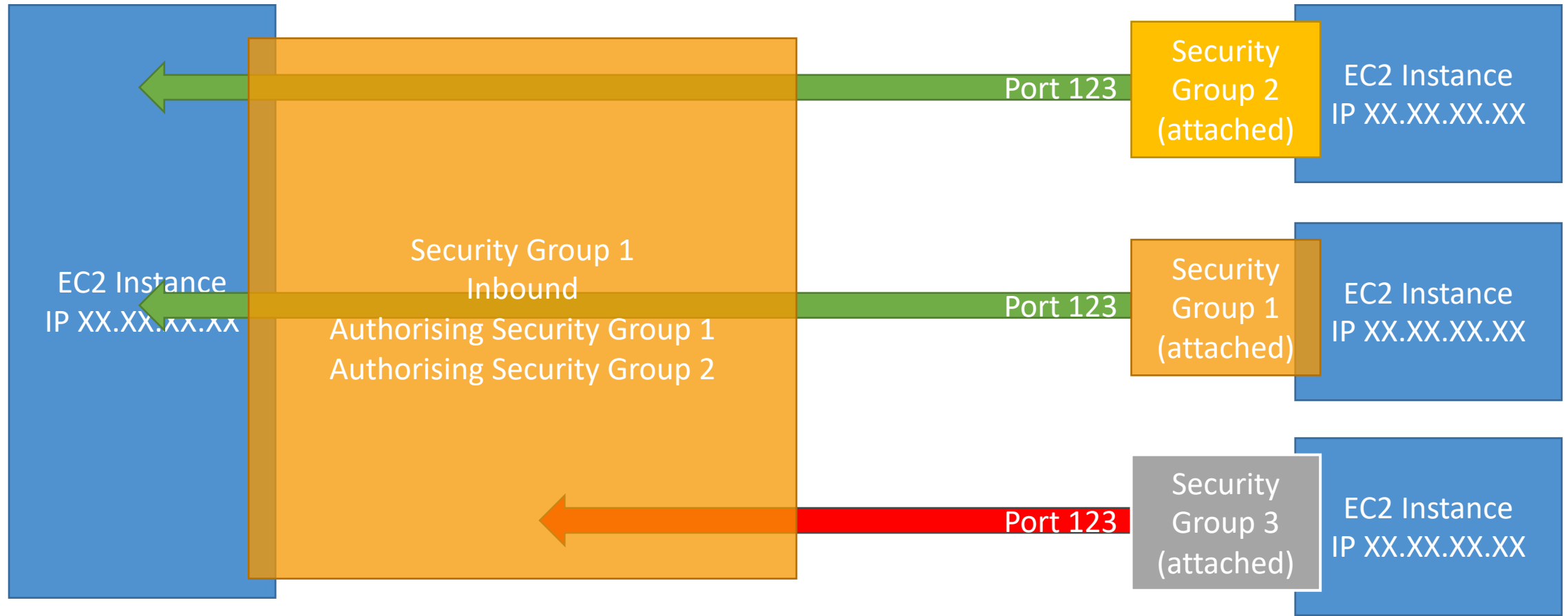
Security Groups

Good to know

- Can be attached to multiple instances
- Locked down to a region / VPC combination
- Does live “outside” the EC2 – if traffic is blocked the EC2 instance won’t see it
- It’s good to maintain one separate security group for SSH access
- If your application is not accessible (time out), then it’s a security group issue
- If your application gives a “connection refused” error, then it’s an application error or it’s not launched
- All inbound traffic is **blocked** by default
- All outbound traffic is **authorised** by default

Referencing other security groups

Diagram



Classic Ports to know

- 22 = SSH (Secure Shell) - log into a Linux instance
- 21 = FTP (File Transfer Protocol) – upload files into a file share
- 22 = SFTP (Secure File Transfer Protocol) – upload files using SSH
- 80 = HTTP – access unsecured websites
- 443 = HTTPS – access secured websites
- 3389 = RDP (Remote Desktop Protocol) – log into a Windows instance

SSH Summary Table

	SSH	Putty	EC2 Instance Connect
Mac	✓		✓
Linux	✓		✓
Windows < 10		✓	✓
Windows >= 10	✓	✓	✓

Which Lectures to watch

- Mac / Linux:
 - SSH on Mac/Linux lecture
- Windows:
 - Putty Lecture
 - If Windows 10: SSH on Windows 10 lecture
- All:
 - EC2 Instance Connect lecture

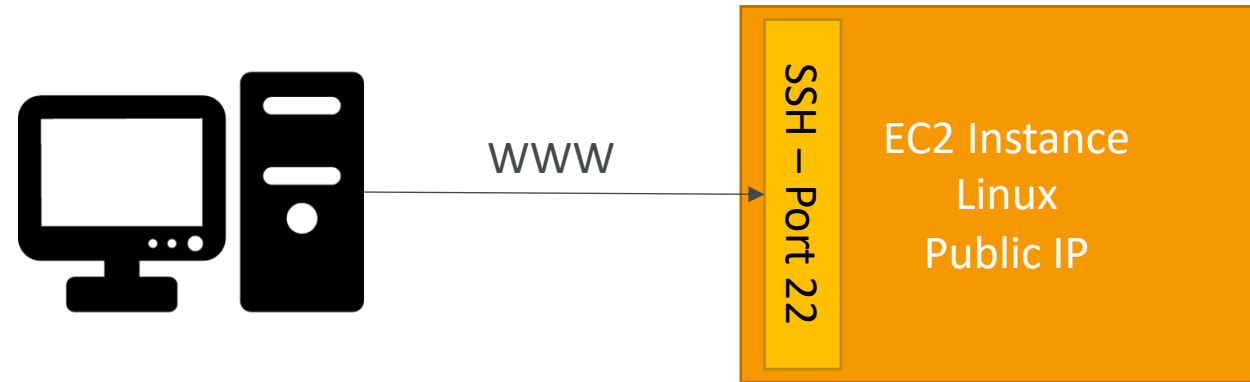
SSH troubleshooting

- Students have the most problems with SSH
- If things don't work...
 1. Re-watch the lecture. You may have missed something
 2. Read the troubleshooting guide
 3. Try EC2 Instance Connect
- If one method works (SSH, Putty or EC2 Instance Connect) you're good
- If no method works, that's okay, the course won't use SSH much

How to SSH into your EC2 Instance

Linux / Mac OS X

- We'll learn how to SSH into your EC2 instance using [Linux / Mac](#)
- SSH is one of the most important function. It allows you to control a remote machine, all using the command line.

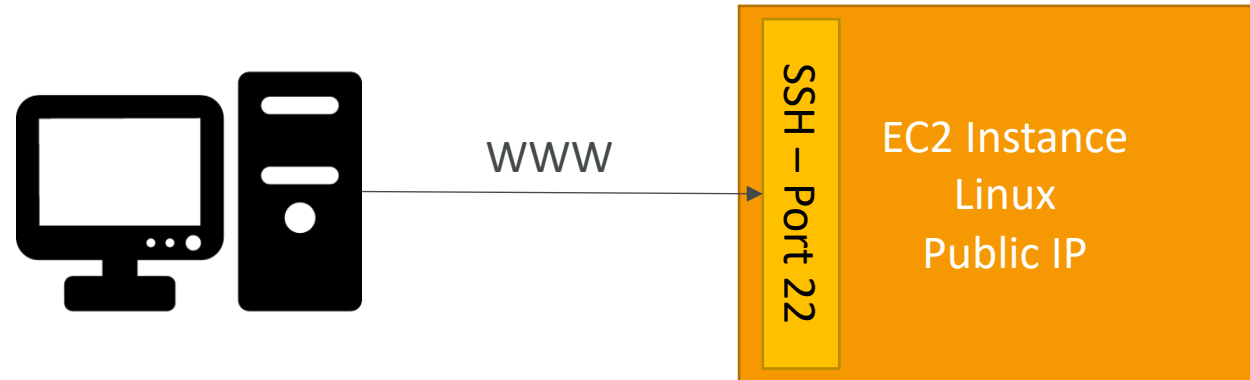


- We will see how we can configure OpenSSH `~/.ssh/config` to facilitate the SSH into our EC2 instances

How to SSH into your EC2 Instance

Windows

- We'll learn how to SSH into your EC2 instance using [Windows](#)
- SSH is one of the most important function. It allows you to control a remote machine, all using the command line.



- We will configure all the required parameters necessary for doing SSH on Windows using the free tool [Putty](#).

EC2 Instance Connect

- Connect to your EC2 instance within your browser
- No need to use your key file that was downloaded
- The “magic” is that a temporary key is uploaded onto EC2 by AWS
- Works only out-of-the-box with Amazon Linux 2
- Need to make sure the port 22 is still opened!

EC2 Instances Purchasing Options

- **On-Demand Instances** – short workload, predictable pricing, pay by second
- **Reserved (1 & 3 years)**
 - **Reserved Instances** – long workloads
 - **Convertible Reserved Instances** – long workloads with flexible instances
- **Savings Plans (1 & 3 years)** – commitment to an amount of usage, long workload
- **Spot Instances** – short workloads, cheap, can lose instances (less reliable)
- **Dedicated Hosts** – book an entire physical server, control instance placement
- **Dedicated Instances** – no other customers will share your hardware
- **Capacity Reservations** – reserve capacity in a specific AZ for any duration

EC2 On Demand

- Pay for what you use:
 - Linux or Windows - billing per second, after the first minute
 - All other operating systems - billing per hour
- Has the highest cost but no upfront payment
- No long-term commitment
- Recommended for **short-term** and **un-interrupted workloads**, where you can't predict how the application will behave

EC2 Reserved Instances

- Up to 72% discount compared to On-demand
- You reserve a specific instance attributes (Instance Type, Region, Tenancy, OS)
- Reservation Period – 1 year (+discount) or 3 years (+++discount)
- Payment Options – No Upfront (+), Partial Upfront (++), All Upfront (+++)
- Reserved Instance's Scope – Regional or Zonal (reserve capacity in an AZ)
- Recommended for steady-state usage applications (think database)
- You can buy and sell in the Reserved Instance Marketplace
- Convertible Reserved Instance
 - Can change the EC2 instance type, instance family, OS, scope and tenancy
 - Up to 66% discount

Note: the % discounts are different from the video as AWS change them over time – the exact numbers are not needed for the exam. This is just for illustrative purposes 😊

EC2 Savings Plans

- Get a discount based on long-term usage (up to 72% - same as RIs)
- Commit to a certain type of usage (\$10/hour for 1 or 3 years)
- Usage beyond EC2 Savings Plans is billed at the On-Demand price
- Locked to a specific instance family & AWS region (e.g., M5 in us-east-1)
- Flexible across:
 - Instance Size (e.g., m5.xlarge, m5.2xlarge)
 - OS (e.g., Linux, Windows)
 - Tenancy (Host, Dedicated, Default)



EC2 Spot Instances

- Can get a **discount of up to 90%** compared to On-demand
- Instances that you can “lose” at any point of time if your max price is less than the current spot price
- The **MOST cost-efficient** instances in AWS
- Useful for workloads that are resilient to failure
 - Batch jobs
 - Data analysis
 - Image processing
 - Any **distributed** workloads
 - Workloads with a flexible start and end time
- Not suitable for critical jobs or databases

EC2 Dedicated Hosts

- A physical server with EC2 instance capacity fully dedicated to your use
- Allows you address **compliance requirements** and **use your existing server-bound software licenses** (per-socket, per-core, per-VM software licenses)
- Purchasing Options:
 - **On-demand** – pay per second for active Dedicated Host
 - **Reserved** - 1 or 3 years (No Upfront, Partial Upfront, All Upfront)
- The most expensive option
- Useful for software that have complicated licensing model (BYOL – Bring Your Own License)
- Or for companies that have strong regulatory or compliance needs

EC2 Dedicated Instances

- Instances run on hardware that's dedicated to you
- May share hardware with other instances in same account
- No control over instance placement (can move hardware after Stop / Start)

Characteristic	Dedicated Instances	Dedicated Hosts
Enables the use of dedicated physical servers	X	X
Per instance billing (subject to a \$2 per region fee)	X	
Per host billing		X
Visibility of sockets, cores, host ID		X
Affinity between a host and instance		X
Targeted instance placement		X
Automatic instance placement	X	X
Add capacity using an allocation request		X

EC2 Capacity Reservations

- Reserve **On-Demand** instances capacity in a specific AZ for any duration
- You always have access to EC2 capacity when you need it
- **No time commitment** (create/cancel anytime), **no billing discounts**
- Combine with Regional Reserved Instances and Savings Plans to benefit from billing discounts
- You're charged at On-Demand rate whether you run instances or not
- Suitable for short-term, uninterrupted workloads that needs to be in a specific AZ

Which purchasing option is right for me?



- **On demand:** coming and staying in resort whenever we like, we pay the full price
- **Reserved:** like planning ahead and if we plan to stay for a long time, we may get a good discount.
- **Savings Plans:** pay a certain amount per hour for certain period and stay in any room type (e.g., King, Suite, Sea View, ...)
- **Spot instances:** the hotel allows people to bid for the empty rooms and the highest bidder keeps the rooms. You can get kicked out at any time
- **Dedicated Hosts:** We book an entire building of the resort
- **Capacity Reservations:** you book a room for a period with full price even you don't stay in it

Price Comparison

Example – m4.large – us-east-1

Price Type	Price (per hour)
On-Demand	\$0.10
Spot Instance (Spot Price)	\$0.038 - \$0.039 (up to 61% off)
Reserved Instance (1 year)	\$0.062 (No Upfront) - \$0.058 (All Upfront)
Reserved Instance (3 years)	\$0.043 (No Upfront) - \$0.037 (All Upfront)
EC2 Savings Plan (1 year)	\$0.062 (No Upfront) - \$0.058 (All Upfront)
Reserved Convertible Instance (1 year)	\$0.071 (No Upfront) - \$0.066 (All Upfront)
Dedicated Host	On-Demand Price
Dedicated Host Reservation	Up to 70% off
Capacity Reservations	On-Demand Price

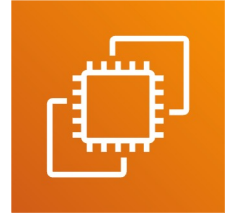
Shared Responsibility Model for EC2



- Infrastructure (global network security)
- Isolation on physical hosts
- Replacing faulty hardware
- Compliance validation



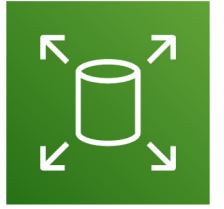
- Security Groups rules
- Operating-system patches and updates
- Software and utilities installed on the EC2 instance
- IAM Roles assigned to EC2 & IAM user access management
- Data security on your instance



EC2 Section – Summary

- **EC2 Instance:** AMI (OS) + Instance Size (CPU + RAM) + Storage + security groups + EC2 User Data
- **Security Groups:** Firewall attached to the EC2 instance
- **EC2 User Data:** Script launched at the first start of an instance
- **SSH:** start a terminal into our EC2 Instances (port 22)
- **EC2 Instance Role:** link to IAM roles
- **Purchasing Options:** On-Demand, Spot, Reserved (Standard + Convertible), Dedicated Host, Dedicated Instance

EC2 Instance Storage Section



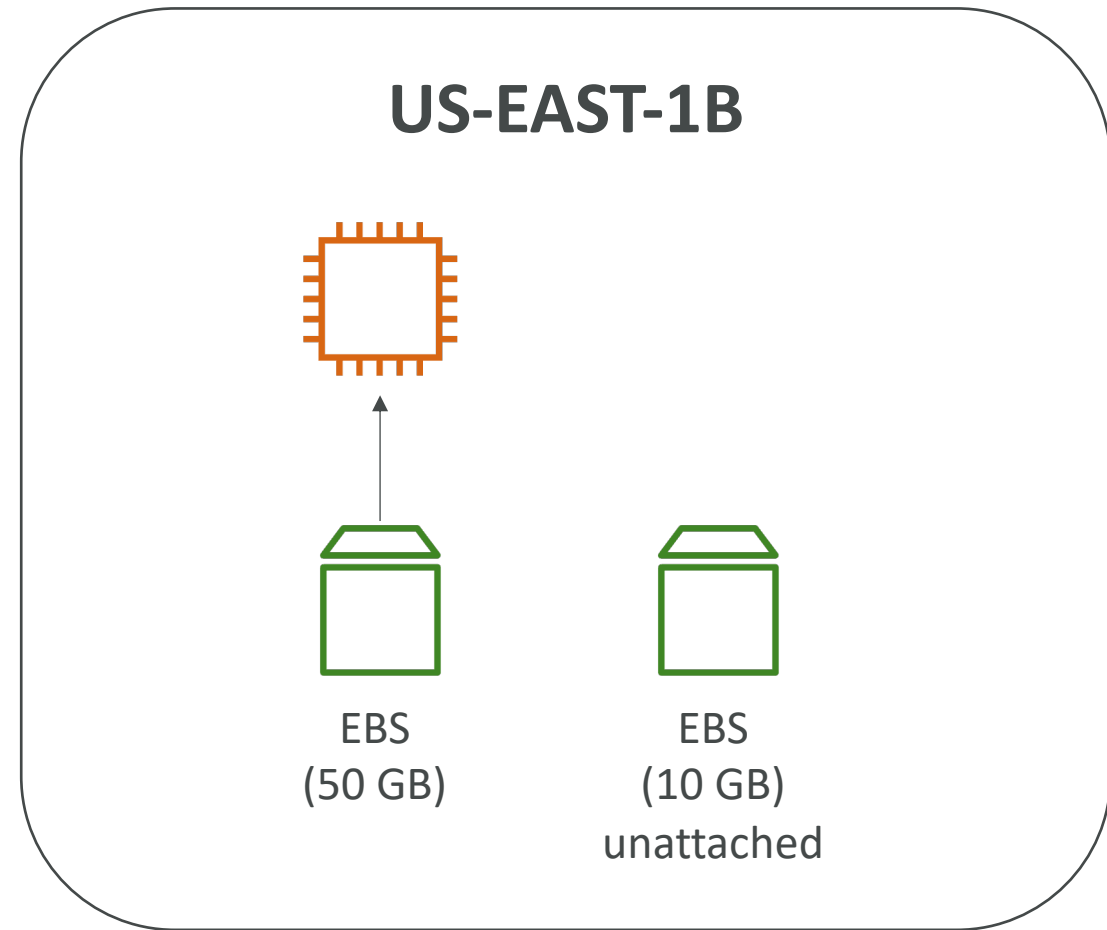
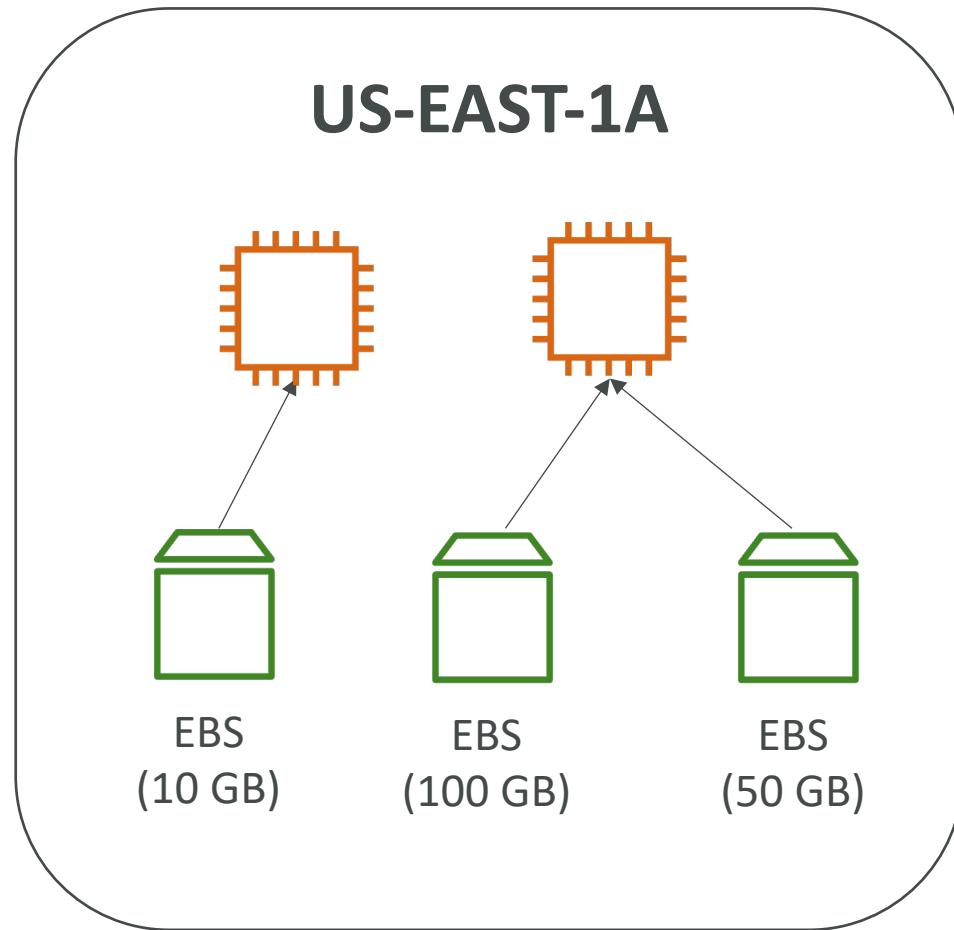
What's an EBS Volume?

- An **EBS (Elastic Block Store) Volume** is a **network** drive you can attach to your instances while they run
- It allows your instances to persist data, even after their termination
- **They can only be mounted to one instance at a time** (at the CCP level)
- They are bound to a **specific availability zone**
- Analogy: Think of them as a “network USB stick”

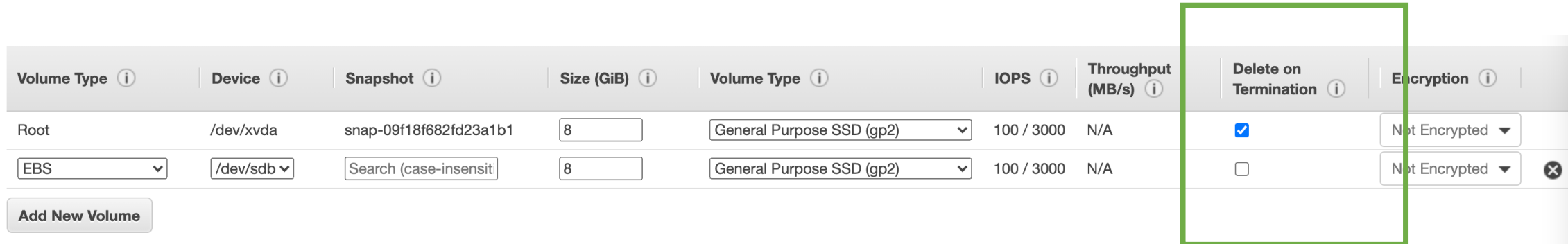
EBS Volume

- It's a network drive (i.e. not a physical drive)
 - It uses the network to communicate the instance, which means there might be a bit of latency
 - It can be detached from an EC2 instance and attached to another one quickly
- It's locked to an Availability Zone (AZ)
 - An EBS Volume in us-east-1a cannot be attached to us-east-1b
 - To move a volume across, you first need to snapshot it
- Have a provisioned capacity (size in GBs, and IOPS)
 - You get billed for all the provisioned capacity
 - You can increase the capacity of the drive over time

EBS Volume - Example



EBS – Delete on Termination attribute



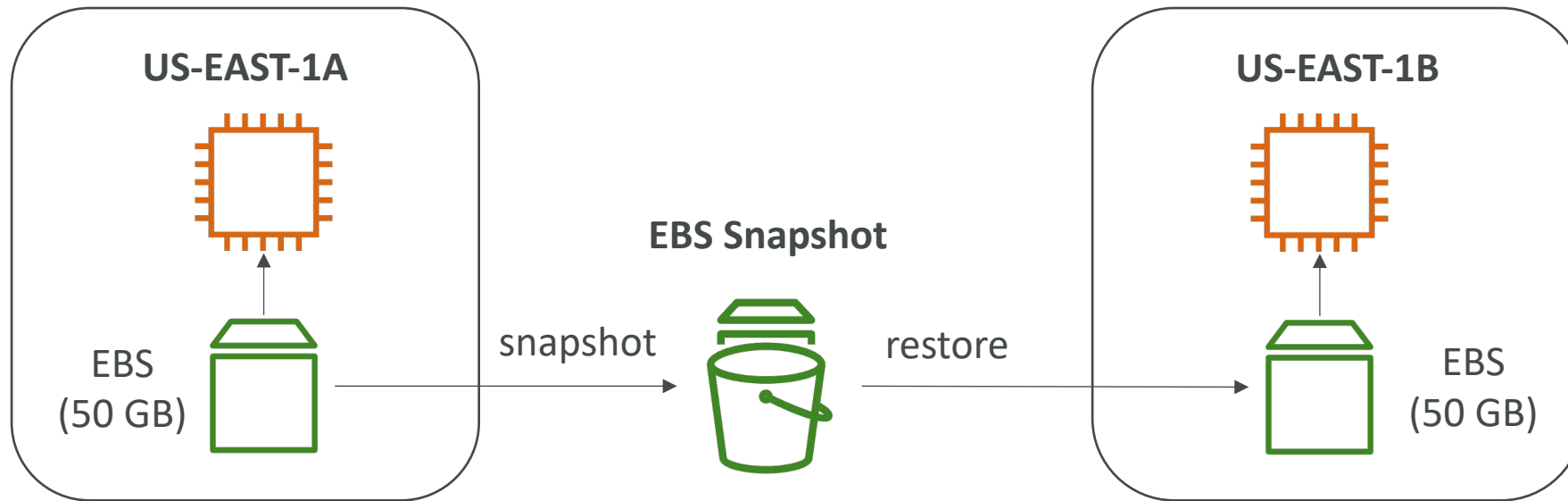
Volume Type ⓘ	Device ⓘ	Snapshot ⓘ	Size (GiB) ⓘ	Volume Type ⓘ	IOPS ⓘ	Throughput (MB/s) ⓘ	Delete on Termination ⓘ	Encryption ⓘ
Root	/dev/xvda	snap-09f18f682fd23a1b1	8	General Purpose SSD (gp2) ▼	100 / 3000	N/A	☒	Not Encrypted ▼
EBS ▼	/dev/sdb ▼	Search (case-insensit	8	General Purpose SSD (gp2) ▼	100 / 3000	N/A	☐	Not Encrypted ▼

Add New Volume

- Controls the EBS behaviour when an EC2 instance terminates
 - By default, the root EBS volume is deleted (attribute enabled)
 - By default, any other attached EBS volume is not deleted (attribute disabled)
- This can be controlled by the AWS console / AWS CLI
- Use case: preserve root volume when instance is terminated

EBS Snapshots

- Make a backup (snapshot) of your EBS volume at a point in time
- Not necessary to detach volume to do snapshot, but recommended
- Can copy snapshots across AZ or Region



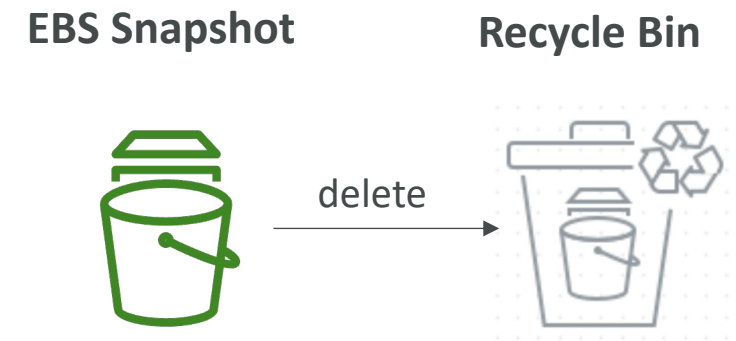
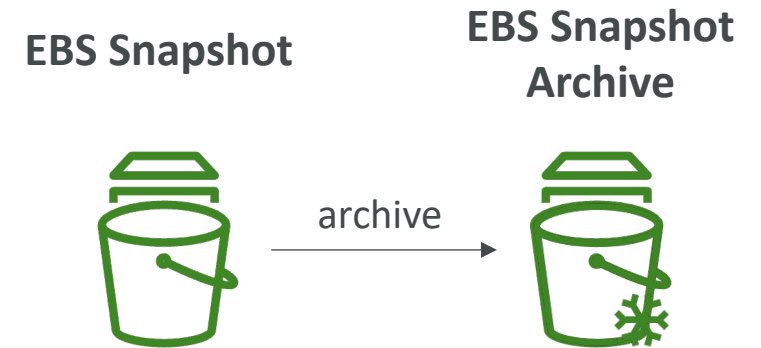
EBS Snapshots Features

- EBS Snapshot Archive

- Move a Snapshot to an "archive tier" that is 75% cheaper
- Takes within 24 to 72 hours for restoring the archive

- Recycle Bin for EBS Snapshots

- Setup rules to retain deleted snapshots so you can recover them after an accidental deletion
- Specify retention (from 1 day to 1 year)



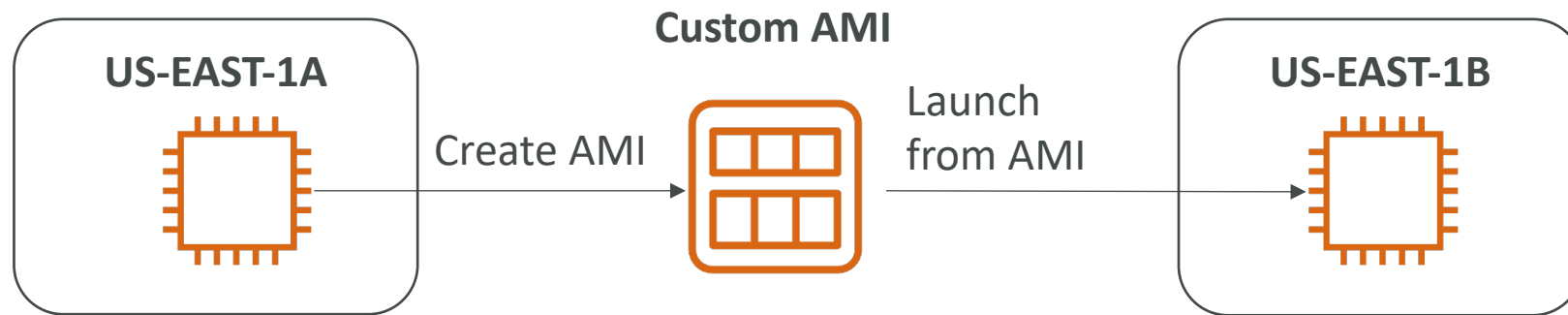


AMI Overview

- AMI = Amazon Machine Image
- AMI are a **customization** of an EC2 instance
 - You add your own software, configuration, operating system, monitoring...
 - Faster boot / configuration time because all your software is pre-packaged
- AMI are built for a **specific region** (and can be copied across regions)
- You can launch EC2 instances from:
 - A Public AMI: AWS provided
 - Your own AMI: you make and maintain them yourself
 - An AWS Marketplace AMI: an AMI someone else made (and potentially sells)

AMI Process (from an EC2 instance)

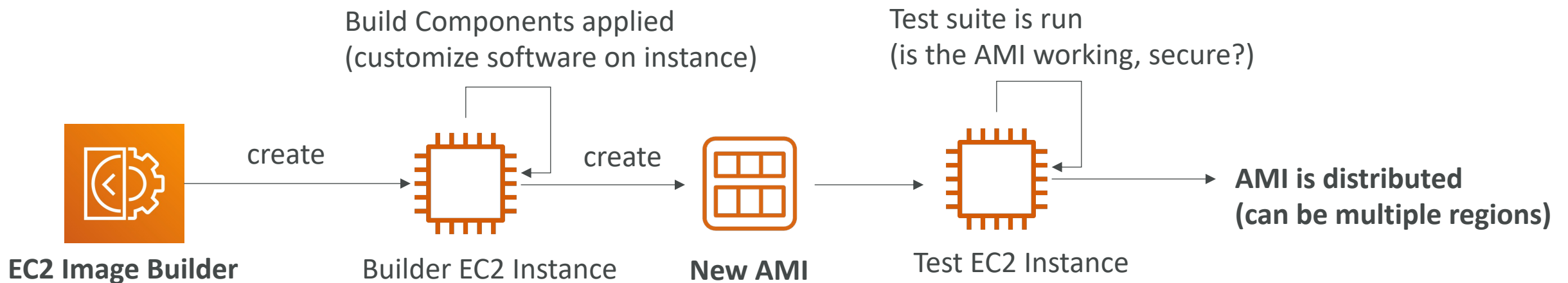
- Start an EC2 instance and customize it
- Stop the instance (for data integrity)
- Build an AMI – this will also create EBS snapshots
- Launch instances from other AMIs

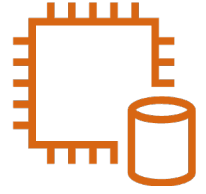




EC2 Image Builder

- Used to automate the creation of Virtual Machines or container images
- => Automate the creation, maintain, validate and test **EC2 AMIs**
- Can be run on a schedule (weekly, whenever packages are updated, etc...)
- Free service (only pay for the underlying resources)





EC2 Instance Store

- EBS volumes are **network drives** with good but “limited” performance
- If you need a high-performance hardware disk, use EC2 Instance Store
- Better I/O performance
- EC2 Instance Store lose their storage if they're stopped (ephemeral)
- Good for buffer / cache / scratch data / temporary content
- Risk of data loss if hardware fails
- Backups and Replication are your responsibility

Local EC2 Instance Store

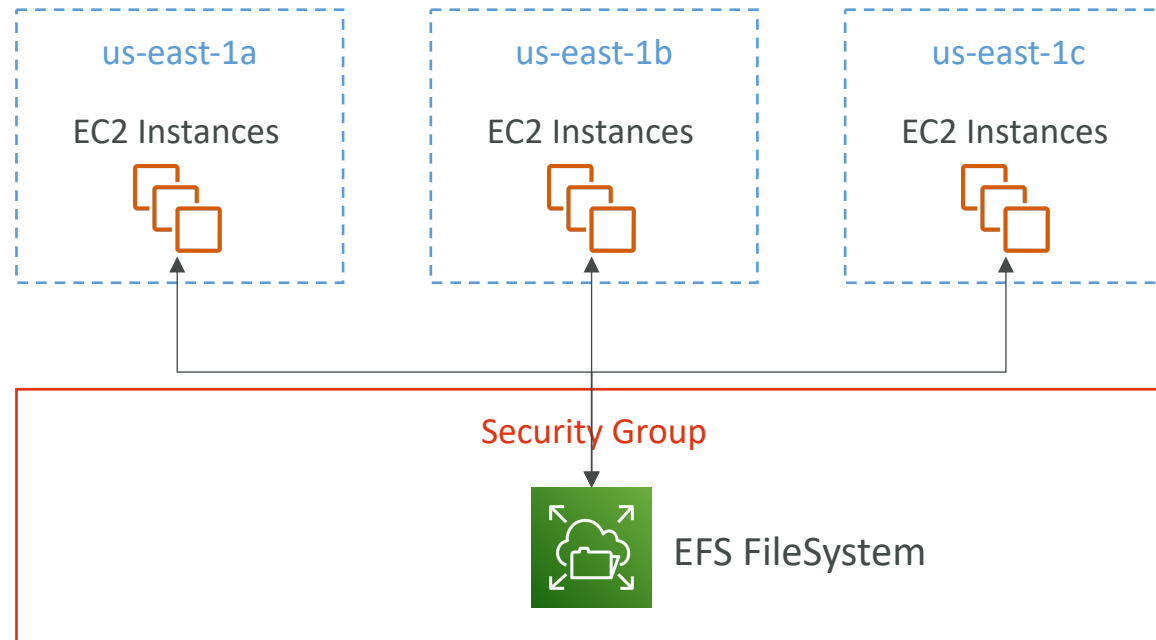
Very high IOPS

Instance Size	100% Random Read IOPS	Write IOPS
i3.large *	100,125	35,000
i3.xlarge *	206,250	70,000
i3.2xlarge	412,500	180,000
i3.4xlarge	825,000	360,000
i3.8xlarge	1.65 million	720,000
i3.16xlarge	3.3 million	1.4 million
i3.metal	3.3 million	1.4 million
i3en.large *	42,500	32,500
i3en.xlarge *	85,000	65,000
i3en.2xlarge *	170,000	130,000
i3en.3xlarge	250,000	200,000
i3en.6xlarge	500,000	400,000
i3en.12xlarge	1 million	800,000
i3en.24xlarge	2 million	1.6 million
i3en.metal	2 million	1.6 million

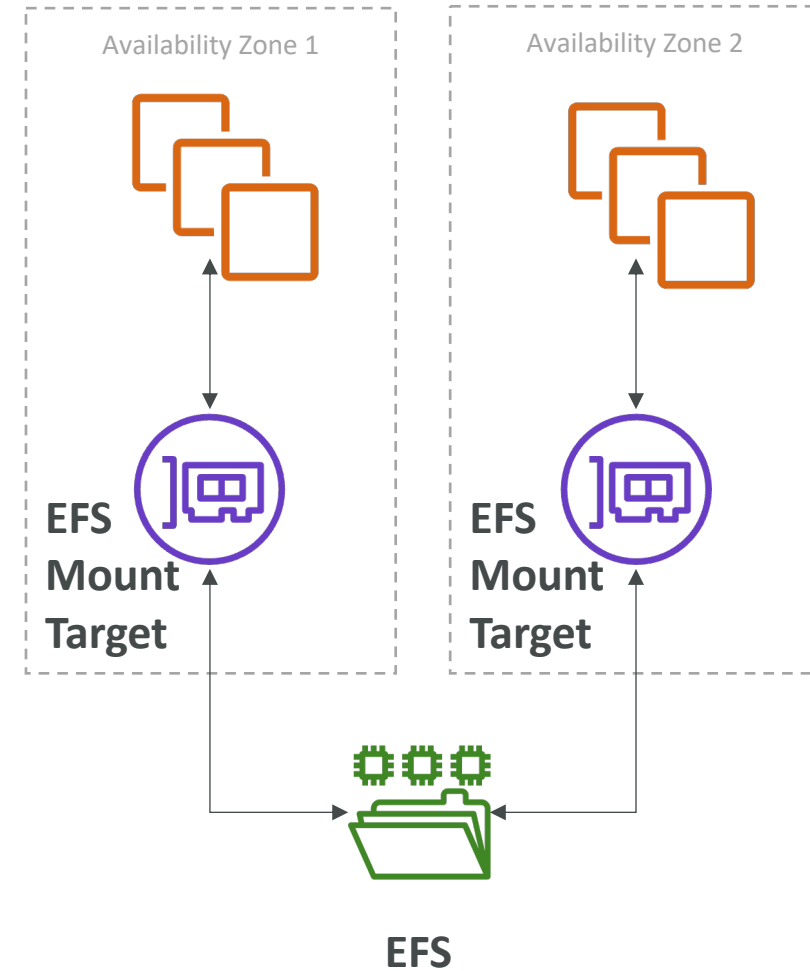
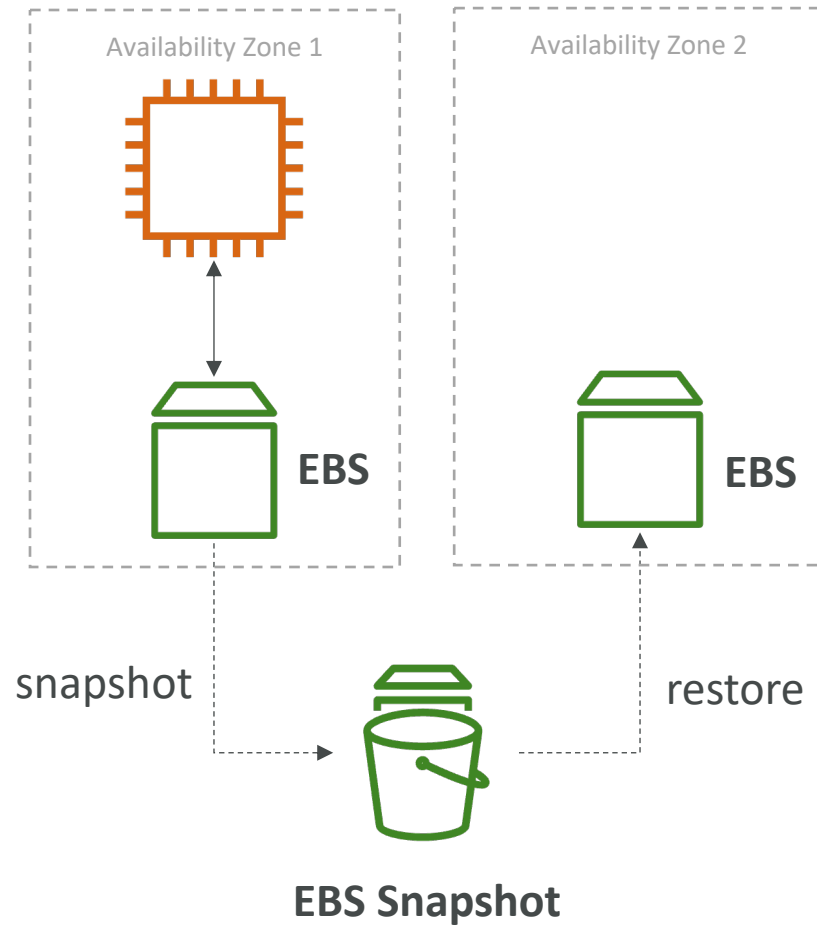
EFS – Elastic File System



- Managed NFS (network file system) that can be mounted on 100s of EC2
- EFS works with **Linux** EC2 instances in **multi-AZ**
- Highly available, scalable, expensive (3x gp2), pay per use, no capacity planning

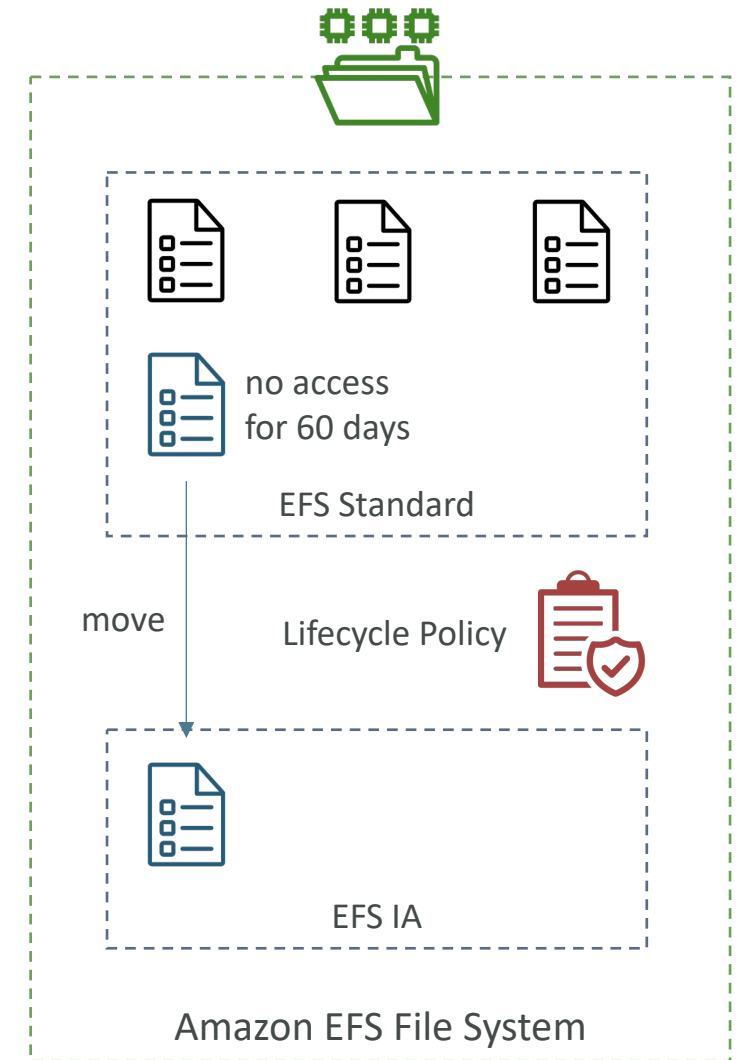


EBS vs EFS



EFS Infrequent Access (EFS-IA)

- **Storage class** that is cost-optimized for files not accessed every day
- Up to 92% lower cost compared to EFS Standard
- EFS will automatically move your files to EFS-IA based on the last time they were accessed
- Enable EFS-IA with a Lifecycle Policy
- Example: move files that are not accessed for 60 days to EFS-IA
- Transparent to the applications accessing EFS



Shared Responsibility Model for EC2 Storage



- Infrastructure
- Replication for data for EBS volumes & EFS drives
- Replacing faulty hardware
- Ensuring their employees cannot access your data

- Setting up backup / snapshot procedures
- Setting up data encryption
- Responsibility of any data on the drives
- Understanding the risk of using EC2 Instance Store



Amazon FSx – Overview

- Launch 3rd party high-performance file systems on AWS
- Fully managed service



FSx for Lustre



**FSx for
Windows File
Server**

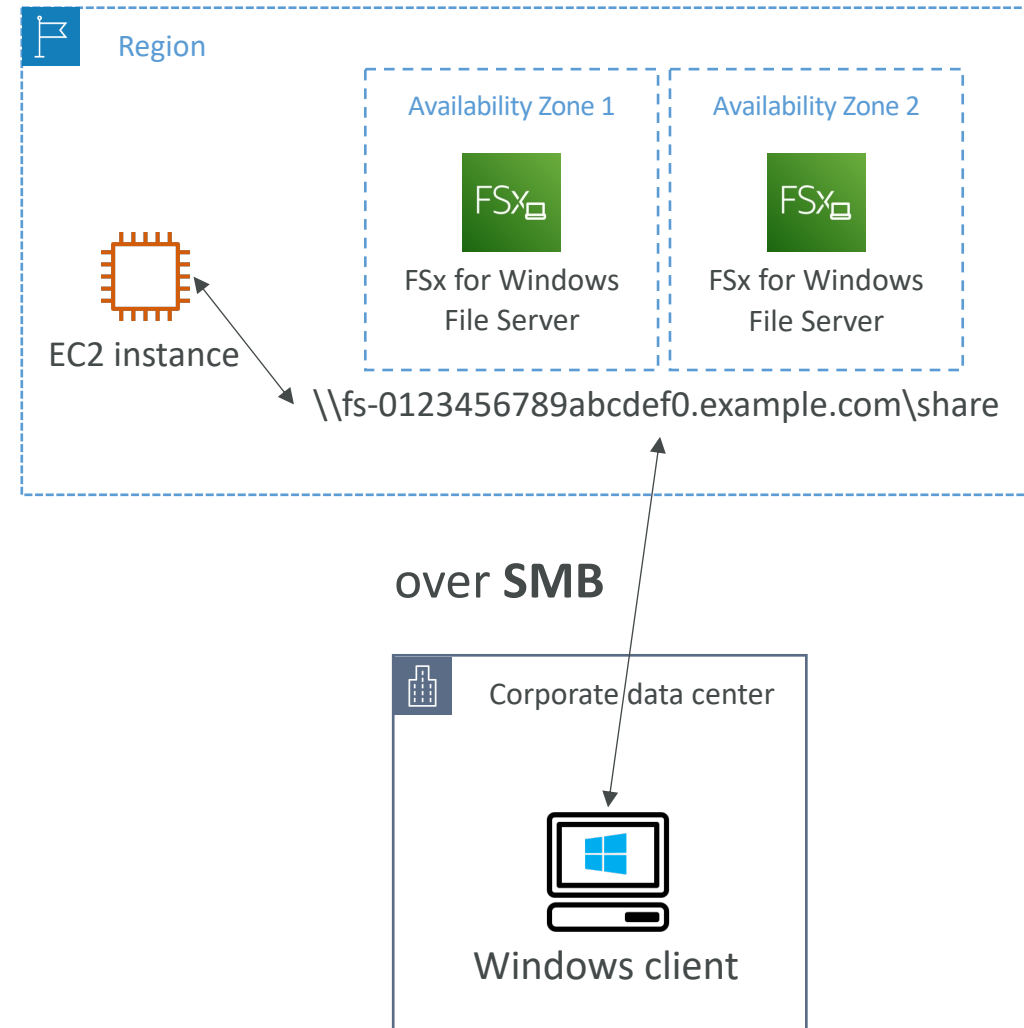


**FSx for
NetApp ONTAP**

Amazon FSx for Windows File Server

FSx

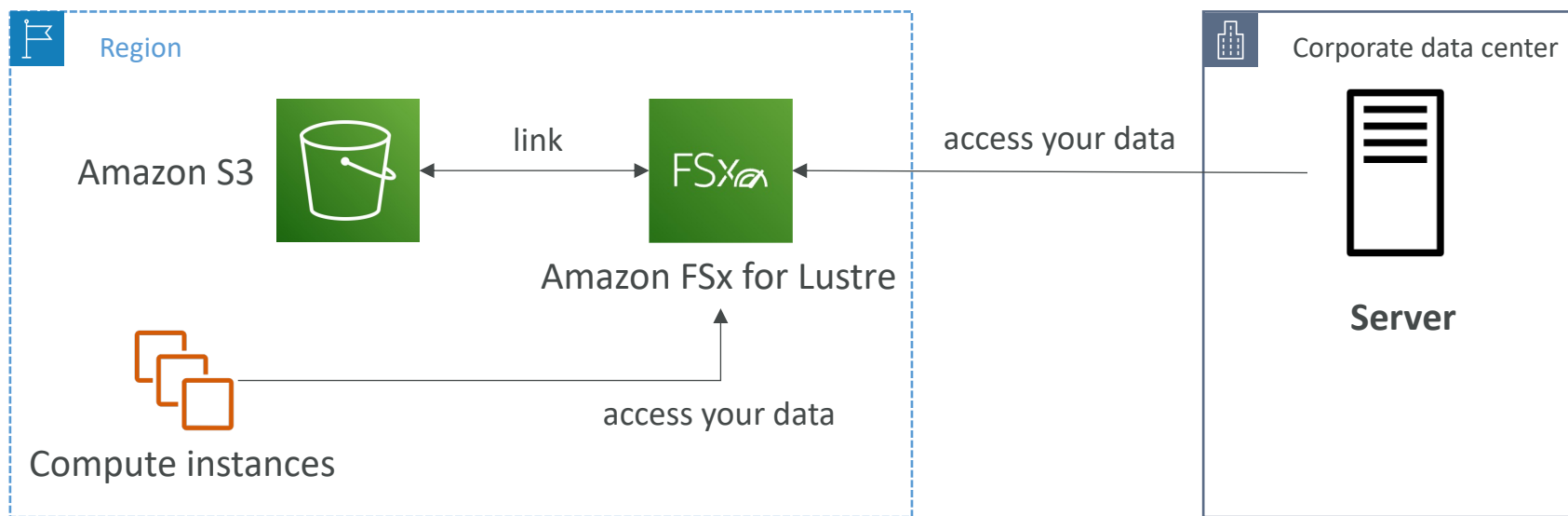
- A fully managed, highly reliable, and scalable **Windows native** shared file system
- Built on **Windows File Server**
- Supports **SMB** protocol & Windows NTFS
- Integrated with Microsoft Active Directory
- Can be accessed from AWS or your on-premise infrastructure





Amazon FSx for Lustre

- A fully managed, high-performance, scalable file storage for **High Performance Computing (HPC)**
- The name Lustre is derived from “Linux” and “cluster”
- Machine Learning, Analytics, Video Processing, Financial Modeling, ...
- Scales up to 100s GB/s, millions of IOPS, sub-ms latencies



EC2 Instance Storage - Summary

- **EBS volumes:**
 - network drives attached to one EC2 instance at a time
 - Mapped to an Availability Zones
 - Can use EBS Snapshots for backups / transferring EBS volumes across AZ
- **AMI:** create ready-to-use EC2 instances with our customizations
- **EC2 Image Builder:** automatically build, test and distribute AMIs
- **EC2 Instance Store:**
 - High performance hardware disk attached to our EC2 instance
 - Lost if our instance is stopped / terminated
- **EFS:** network file system, can be attached to 100s of instances in a region
- **EFS-IA:** cost-optimized storage class for infrequent accessed files
- **FSx for Windows:** Network File System for Windows servers
- **FSx for Lustre:** High Performance Computing Linux file system